Introduction to Database

Semester Project

Pakistan Census Database

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18. Documentation (10 Marks)

Include a PDF or Word document containing:

- **Project Title**
- **Group Members and Roles**
- **ERD and Relational Mapping**
- **SQL Queries with Outputs**
- **Screenshots**
- **Explanation of Logic**
- **Challenges Faced**
- **Conclusion**

1. Project Title

Pakistan Census Database

2. Project Overview

The **Pakistan Census Database** is a centralized system built to collect and manage national demographic, household, and geographic data.

- **Geographic Tracking:** Organizes data down a strict chain from Provinces and Districts to local Census Blocks.
- **Property & Housing Utilities:** Tracks the quality of physical structures and access to water, power, and fuel.
- **Population Registries:** Stores citizen profiles, including CNICs, family structures, literacy, and income.
- **Field Operations:** Manages the census workforce by linking local enumerators to their regional supervisors.

Ultimately, this system provides fast, reliable data slicing to help the government make accurate decisions regarding resource allocation and national planning.

3. Objectives

Objective 1: Securely store and quickly find data across all geographic levels, from whole provinces down to local census blocks.

Objective 2: Track citizen details accurately, including CNICs, family structures, income, literacy, and housing conditions.

Objective 3: Support generating clear population reports, literacy rate summaries, and regional development statistics for government planning.

Objective 4: Perform complex SQL queries to filter and analyze deep demographic trends and employment rates across different regions.

Objective 5: Ensure database normalization and strict integrity constraints so that data remains clean, connected, and free of duplicates.

4. Scope

What is Included:

- **Geographic & Personnel Infrastructure:** Tracking administrative boundaries from provinces down to local census blocks, along with the operational field staff hierarchies.
- **Property & Household Metrics:** Managing details of physical structures, domestic utility profiles, and family household groupings.
- **Citizen Demographics:** Storing individual citizen records, including identity verification numbers (CNICs), family relations, educational status, and income levels.
- **Backend Database Optimization:** Developing index structures, relational constraints, and core SQL queries for backend data analysis.

What is Not Included:

- **Front-End Applica tions:** The project does not include building web or mobile applications for data entry or public dashboards.
- **Live GPS Tracking:** It maps static coordinates for structures but does not support real-time GPS tracking of field staff.
- **Biometric Verification System:** The database processes text-based CNIC records but does not handle direct fingerprint or facial biometric verification processing.

5. Entities and Attributes

Entity	Attributes
Water_Source	**water_source_id**, source_name
Lighting_Source	**lighting_source_id**, source_name
Cooking_Fuel_Source	**fuel_source_id**, fuel_source_name
Washroom_Status	**washroom_status_id**, status_name
Kitchen_Status	**kitchen_status_id**, status_name
Biological_Sex	**sex_id**, sex_name
Religion	**religion_id**, religion_name
Mother_Tongue	**mother_tongue_id**, tongue_name
Education_Level	**education_level_id**, level_name
Employment_Status	**employment_status_id**, status_name
Marital_Status	**marital_status_id**, status_name
Disability_Type	**disability_type_id**, disability_name
Occupation	**occupation_id**, occupation_name
Industry	**industry_id**, industry_name
Relationship_Type	**relationship_type_id**, relationship_name

Entity	Attributes
Census_Cycle	**census_id**, census_year, census_title, start_date, end_date
Census_Staff	**staff_id**, first_name, last_name, staff_role, contact_number, email, assigned_census_block_id, supervisor_id, employment_date

Entity	Attributes
Nationality	**nationality_id**, country_name, citizenship_status
Province	**province_id**, province_name
Division	**division_id**, division_name, province_id
District	**district_id**, district_name, division_id
Tehsil	**tehsil_id**, tehsil_name, district_id
Union_Council	**union_council_id**, union_council_name, union_council_classification_id, tehsil_id
Census_Block	**census_block_id**, census_block_code, estimated_household_count, is_urban, area_sq_km, union_council_id
Union_Council_Classification	**union_council_classification_id**, classification_name, description

Entity	Attributes
Person	**person_id**, household_id, first_name, last_name, cnic_number, sex_id, date_of_birth, nationality_id, religion_id, mother_tongue_id, marital_status_id, relationship_type_id, head_person_id, is_literate, education_level_id, currently_attending_school, employment_status_id, occupation_id, industry_id, monthly_income, data_entry_operator_id, created_date
Person_Disability	**person_disability_id**, **person_id**, disability_type_id
Migration_History	**migration_id**, **person_id**, previous_district_id, migration_reason, year_of_migration

Entity	Attributes
Structure_Type	**structure_type_id**, type_name, description
Structure	**structure_id**, structure_address_no, gps_latitude, gps_longitude, created_date, census_block_id, structure_type_id, water_source_id, lighting_source_id, fuel_source_id, washroom_status_id, kitchen_status_id
Structure_Residential	**structure_id**, total_residential_units
Structure_Commercial	**structure_id**, commercial_type_id, business_name, estimated_active_employees, has_industrial_power_connection, has_active_license
Structure_Institution	**structure_id**, institution_type, organization_name, population_count, is_government
Commercial_Sector_Type	**commercial_type_id**, commercial_type_name, description
Household	**household_id**, structure_id, census_id, household_sub_token, household_category_id, residential_status_type_id, owner_sex_id, number_of_rooms, household_size, data_entry_operator_id, created_date
Household_Category	**household_category_id**, category_name, description
Residential_Status_Type	**residential_status_type_id**, status_name, description

6. Relationships

Geographic Hierarchy Chain (One-to-Many Loops)

Province to Division:

- ***Cardinality:*** **1:M** (One Province can have Many Divisions).
- ***Participation:*** **Total** on Division side (A division cannot exist without being mapped to a province).
- ***Foreign Key:*** Division.province_id references Province.province_id.

Division to District:

- ***Cardinality:*** **1:M** (One Division contains Many Districts).
- ***Participation:*** **Total** on District side (A district must belong to an administrative division).
- ***Foreign Key:*** District.division_id references Division.division_id.

District to Tehsil:

- ***Cardinality:*** **1:M** (One District oversees Many Tehsils).
- ***Participation:*** **Total** on Tehsil side.
- ***Foreign Key:*** Tehsil.district_id references District.district_id.

Tehsil to Union Council:

- ***Cardinality:*** **1:M** (One Tehsil contains Many Union Councils).
- ***Participation:*** **Total** on Union Council side.
- ***Foreign Key:*** Union_Council.tehsil_id references Tehsil.tehsil_id.

Union Council to Census Block:

- ***Cardinality:*** **1:M** (One Union Council contains Many localized Census Blocks).
- ***Participation:*** **Total** on Census Block side.
- ***Foreign Key:*** Census_Block.union_council_id references Union_Council.union_council_id.

Specialized Building Extension Layers (1:1 Class Inheritance)

Structure to Structure_Residential:

- ***Cardinality:*** **1:1**
- ***Participation:*** **Partial** on Structure side, **Total** on Residential side (The profile cannot exist without its base parent building asset).
- ***Foreign Key:*** Structure_Residential.structure_id references Structure.structure_id.

Structure to Structure_Commercial:

- ***Cardinality:*** **1:1**
- ***Participation:*** **Partial** on Structure side, **Total** on Commercial side.
- ***Foreign Key:*** Structure_Commercial.structure_id references Structure.structure_id.

Structure to Structure_Institution:

- ***Cardinality:*** **1:1**.
- ***Participation:*** **Partial** on Structure side, **Total** on Institution side.
- ***Foreign Key:*** Structure_Institution.structure_id references Structure.structure_id.

• Customer places Order (One-to-Many)
• Order contains Book (Many-to-Many) — requires a junction table like OrderDetails
• Employee processes Order (One-to-Many)

Core Operational Census Operations

Census_Block to Structure Location:

- ***Cardinality:*** **1:M** (One Census Block encompasses Many physical building structures).
- ***Participation:*** **Total** on Structure side (Every building must be cataloged inside an active census block boundary).
- ***Foreign Key:*** Structure.census_block_id references Census_Block.census_block_id.

Structure Location to Household Unit:

- ***Cardinality:*** **1:M** (A physical building structure can host Multiple separate living households).
- ***Participation:*** **Total** on Household side (Every household must be linked to a physical address/structure asset).
- ***Foreign Key:*** Household.structure_id references Structure.structure_id.

Household Unit to Person Profile:

- ***Cardinality:*** **1:M** (One Household unit contains Many citizens living/eating together).
- ***Participation:*** **Total** on Person side (An individual citizen must be accounted for within an registered household cohort).
- ***Foreign Key:*** Person.household_id references Household.household_id.

Recursive Self-Referencing Management Links

Census_Staff to Census_Staff (Management Hierarchy):

- ***Cardinality:*** **1:M** (One Supervisor manages Many field staff members/enumerators).
- ***Participation:*** **Total** on enumerators side (Enumerators have supervisors).
- ***Foreign Key:*** Census_Staff.supervisor_id references Census_Staff.staff_id.

Person to Person (Household Head Mapping):

- ***Cardinality:*** **1:M** (One household head is referenced by Many living family members in the house).
- ***Participation:*** **Partial** on both sides (A chosen household head points to null or themselves, while family members point to the head).
- ***Foreign Key:*** Person.head_person_id references Person.person_id.

Many-to-Many Dynamic Junction Entities

Person to Disability_Type via Person_Disability:

- ***Cardinality:*** **M:N** (One person can have multiple concurrent functional health challenges; one disability type applies to many citizens).
- ***Participation:*** **Total** on the junction table (Person_Disability), **Partial** on the individual tables (A citizen might have zero disabilities listed).
- ***Foreign Keys:*** Person_Disability.person_id references Person.person_id. Person_Disability.disability_type_id references Disability_Type.disability_type_id.

Person to District via Migration_History:

- ***Cardinality:*** **M:N** (A citizen can move through multiple locations over time; a single district acts as a historical home for many citizens).
- ***Participation:*** **Total** on the history tracker (Migration_History), **Partial** on individual tables.
- ***Foreign Keys:*** Migration_History.person_id references Person.person_id. Migration_History.previous_district_id references District.district_id.

7. ERD (Entity Relationship Diagram)

Students must:
• Draw a complete ERD of their project
• Mention primary keys and foreign keys properly
• Clearly show cardinality and participation constraints
• Use proper notation and naming consistency throughout the diagram

8. Relational Mapping / Schema Design

Students must convert the ERD into Relational Schema.

Example Format:

- Customer(CustomerID, Name, Email, Phone)
- Order(OrderID, CustomerID, Date, Total)
- Book(BookID, Title, Genre, Price)

Requirements:
- Mention Primary Keys and Foreign Keys clearly
- Maintain naming consistency throughout the schema
- Explain foreign key references properly

9. DDL Operations

Students must perform all DDL tasks in their projects, including:

- CREATE DATABASE
- CREATE TABLE
- ALTER TABLE
- DROP TABLE
- TRUNCATE TABLE
- RENAME TABLE
- ADD / MODIFY / DROP Constraints

Constraints should include:
- PRIMARY KEY
- FOREIGN KEY
- UNIQUE
- NOT NULL
- CHECK
- DEFAULT

10. DML Operations

Students must perform DML operations on project tables:

- INSERT statements
- UPDATE statements
- DELETE statements

Requirements:
- Add at least 10 realistic records in each table
- Use meaningful and realistic sample data
- Manipulate data using different fields and conditions

11. SQL Queries and Data Retrieval

A. Joins

Students must perform the following joins:

• INNER JOIN
• LEFT JOIN
• RIGHT JOIN
• FULL OUTER JOIN
• NATURAL JOIN
• SELF JOIN

Requirements:
• Perform complex joins involving 3–4 tables
• Explain query logic and output

B. Nested and Correlated Queries

Students must write at least 5 nested and correlated queries using:

Nested Queries:
• IN
• ANY
• ALL
• NOT IN
• NOT ANY
• NOT ALL

Correlated Queries:
• EXISTS
• NOT EXISTS

Requirements:
• Explain the purpose of each query
• Show output with proper explanation
• Perform dry run manually step by step

C. Aggregate Queries

Students must perform aggregate queries using:

• COUNT()
• SUM()

- AVG()
- MIN()
- MAX()
- GROUP BY
- HAVING

Requirements:
- Use project data for meaningful analysis
- Explain query results properly

D. Set Operations

Students must perform:

- UNION
- UNION ALL
- INTERSECT
- EXCEPT / MINUS

Requirements:
- Explain the purpose and output of each operation

12. Views

Students must create at least three VIEWs.

Requirements:
- Explain practical use of each VIEW
- Show SQL code and output
- Demonstrate how VIEWs simplify data retrieval

13 Cascading Operations

Students must demonstrate cascading operations using examples from their schema.

Operations to include:
- ON DELETE CASCADE
- ON DELETE SET NULL

Requirements:
- Explain behavior before and after deletion
- Show practical examples using project tables

14. Database Design Diagram

(Include ERD and Schema Diagram here if possible)

Students should include:
• ER Diagram
• Relational Schema Diagram
• Table Relationships
• Constraint Representation

15. Challenges and Risks

• Handling large volumes of data efficiently
• Ensuring referential integrity during updates and deletions
• Managing concurrent access to data
• Maintaining normalization and avoiding redundancy
• Handling complex query execution and optimization

16. Error Handling and Screenshots

Students must include screenshots of:

• Successful query execution
• Errors encountered during implementation
• Explanation of why the error occurred
• Steps taken to resolve the error

17. Dry Run of Queries

Students must manually trace query execution step by step.
Requirements:
• Explain input conditions
• Show intermediate steps
• Explain final output logically

18. Documentation (10 Marks)

Include a PDF or Word document containing:

• Project Title
• Group Members and Roles
• ERD and Relational Mapping

- SQL Queries with Outputs
- Screenshots
- Explanation of Logic
- Challenges Faced
- Conclusion

19. Video Demonstration & Viva (10 Marks)

Each student must:
- Record a short video showing the project execution
- Explain their assigned role and logic
- Demonstrate SQL queries and outputs
- Prepare for viva questions related to database concepts and implementation

20. Final Submission Must Include

- .sql code files
- Documentation (PDF or Word)
- Screen recording of execution
- ERD and Relational Schema
- Query outputs and screenshots